

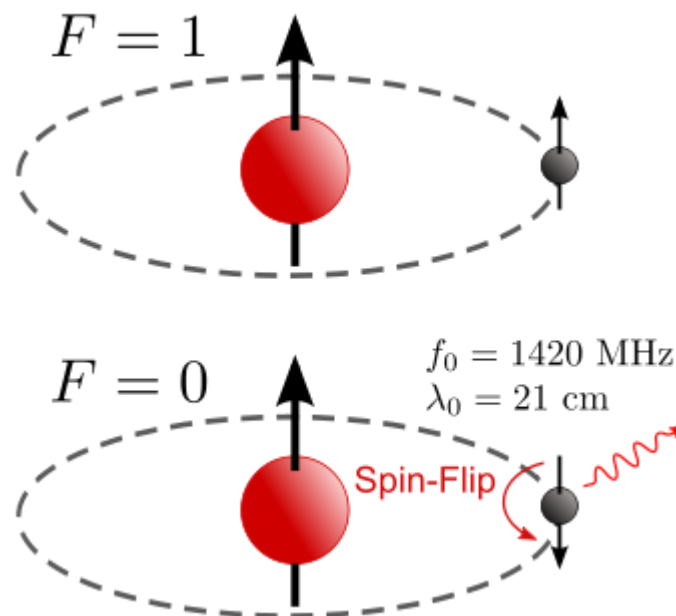
Your Observations

What are we measuring?

Your telescopes are sensitive to neutral hydrogen emission. We call this “HI”.

HI emits at 1420MHz, or 21cm. It is one of the brightest lines in the Universe, and was the first spectral line measured in outer space.

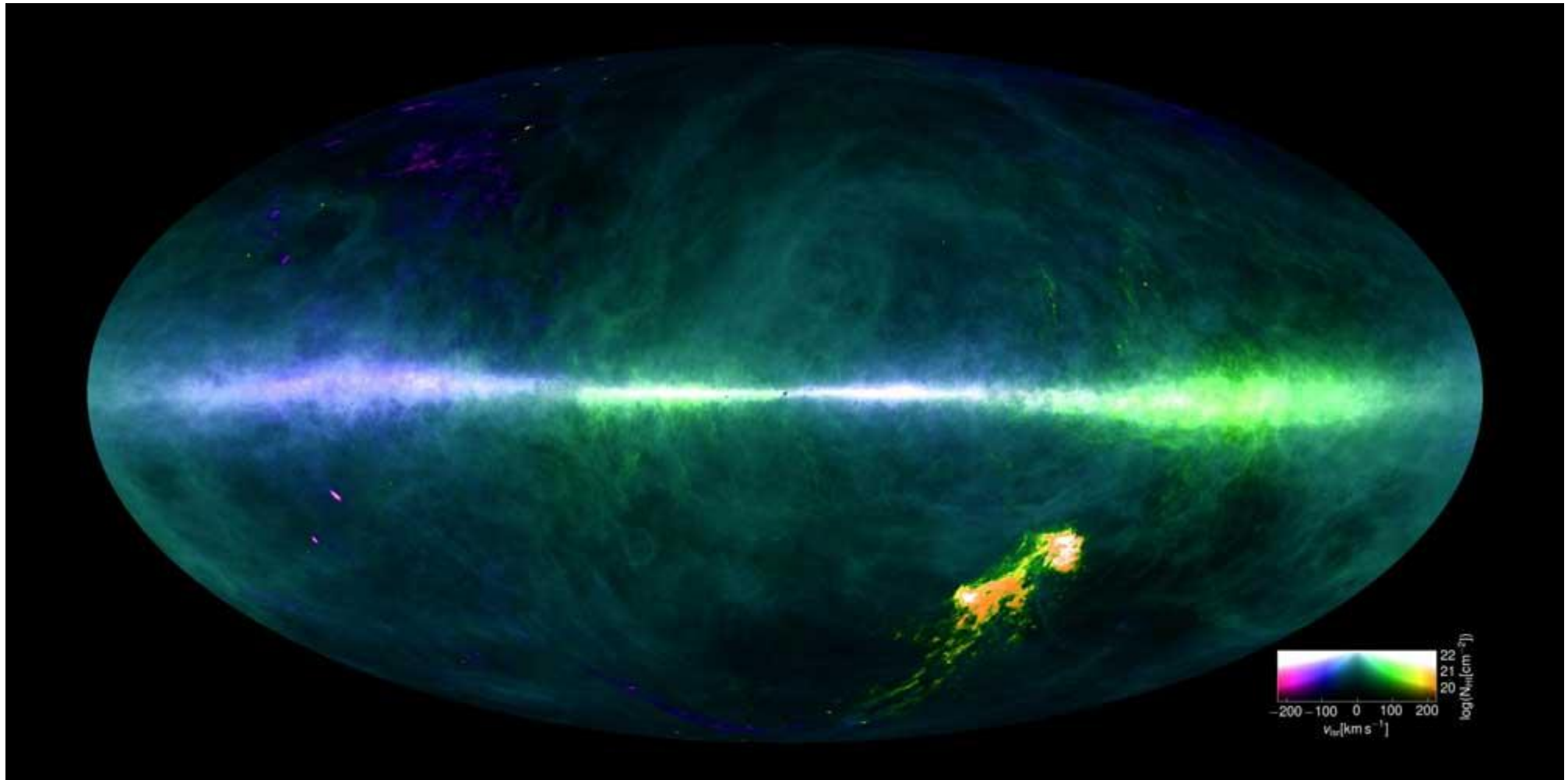
The 21cm line is due to a “spin-flip” transition. The electron and proton’s spin can be aligned or antialigned. This is incredibly rare, but the line is bright because there is so much HI.



Why are we Measuring HI?

HI has two main observational benefits:

- 1) The 21cm line is at a frequency where electronics can be built relatively easily, and
- 2) there is HI everywhere! It can be observed in every direction on the sky.



How Should we Observe?

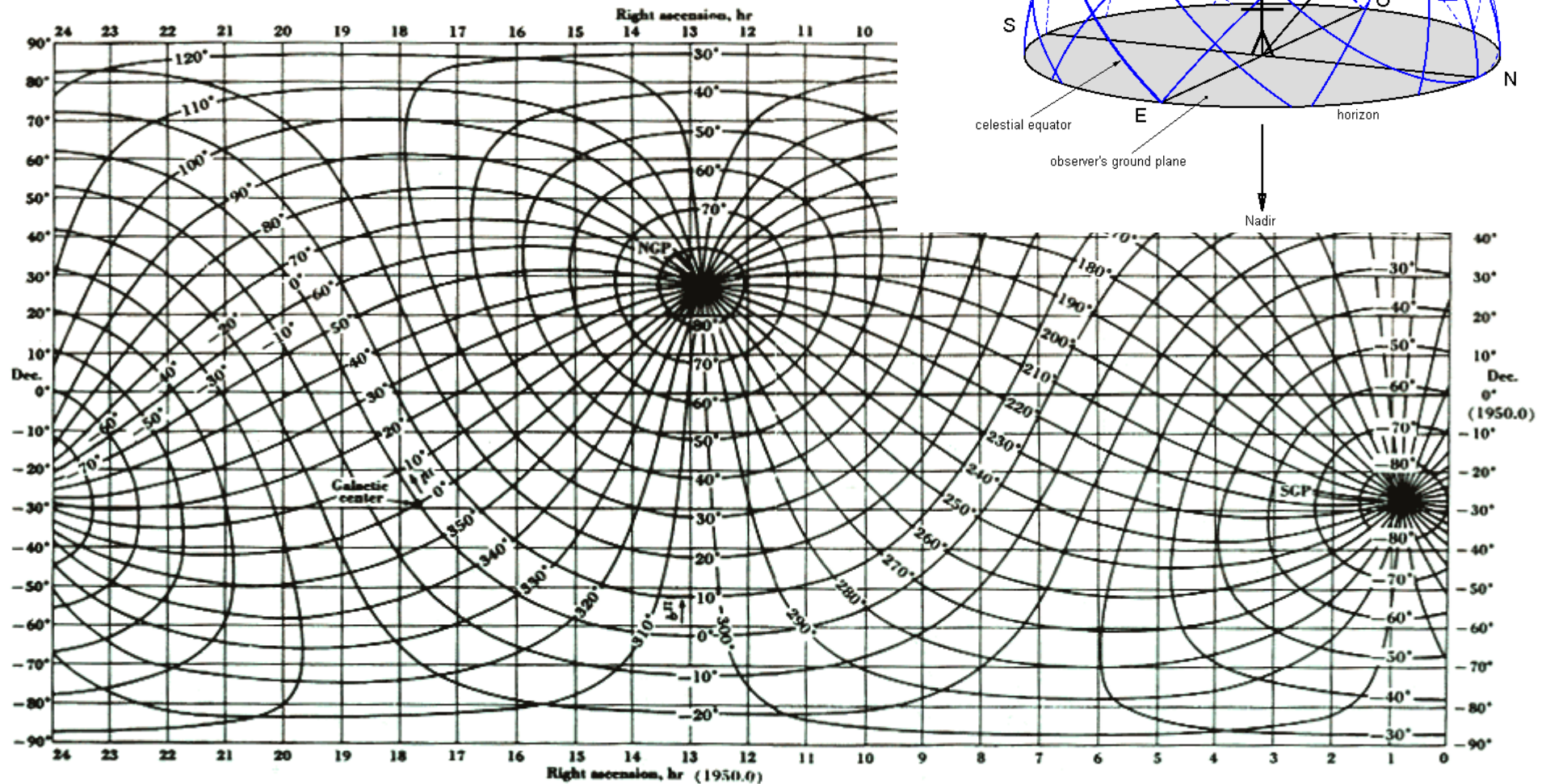
I don't know! There are a couple ways:

- 1) Take data continuously as the sky passes overhead, or
- 2) Target a specific part of the sky

Drift Scans

Taking data continuously as the sky passes overhead is called a “drift scan.”

If the telescope is pointed in one location, you will get data at one dec. but all RAs.



How to do Drift Scans

The easy way:

- 1) Point the telescope somewhere and collect data for as long as you want

The harder way:

- 1) Pick a portion of the Galaxy you want to observe and determine its RA and Dec using
 - a) the graph from last slide or
 - b) this converter: heasarc.nasa.gov/cgi-bin/Tools/convcoord/convcoord.pl
- 2) Determine the current LST and use this to find when the source transits. Find the LST with:
 - a) the Sun (very approximate) or
 - b) this calculator <http://www.wvu.edu/skywise/skymobile/skywatch.html>
- 3) Point your telescope and collect data! You'll need to determine the altitude using:
 - a) the fact that the altitude of Dec = 90deg is your latitude on Earth or
 - b) this equation: $\sin(\text{alt}) = \sin(\text{Dec}) \sin(\text{lat}) + \cos(\text{Dec}) \cos(\text{HA}) \cos(\text{lat})$, with HA = 0. For Lat = 39deg.,
 $\text{alt} = \sin^{-1} (0.63 \sin(\text{Dec}) + 0.78 \cos(\text{Dec}))$

Pointed Observations

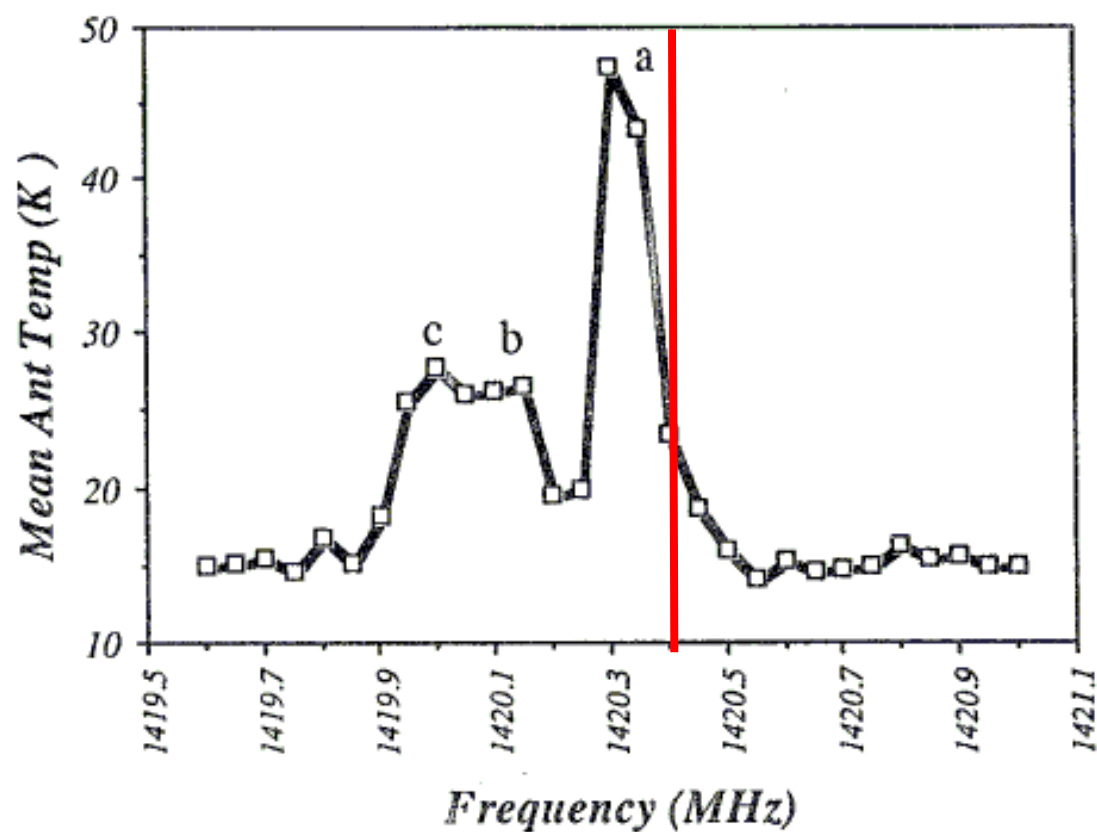
The is the same as the harder drift scan method. If there is a particular portion of the sky you want to observe:

- 1) Determine its longitude and latitude and convert that to RA and Dec using
 - a) the graph from last slide or
 - b) this converter: heasarc.nasa.gov/cgi-bin/Tools/convcoord/convcoord.pl
- 2) Determine the current LST and use this to find when the source transits. Find the LST with:
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Turning into Velocities

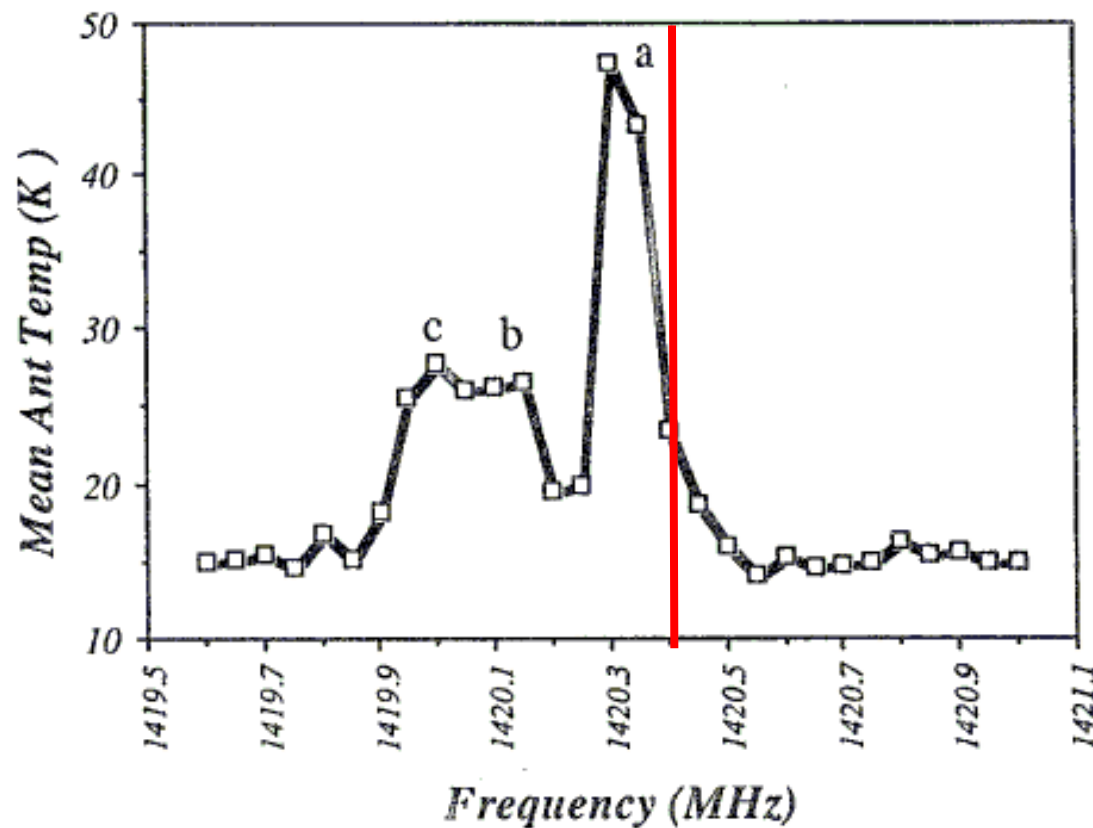
Now you have data! YAY! Let's put it into velocity using the Doppler correction:

$$v = (v_0 - \nu)c/\nu_0 = \Delta\nu c/\nu_0$$



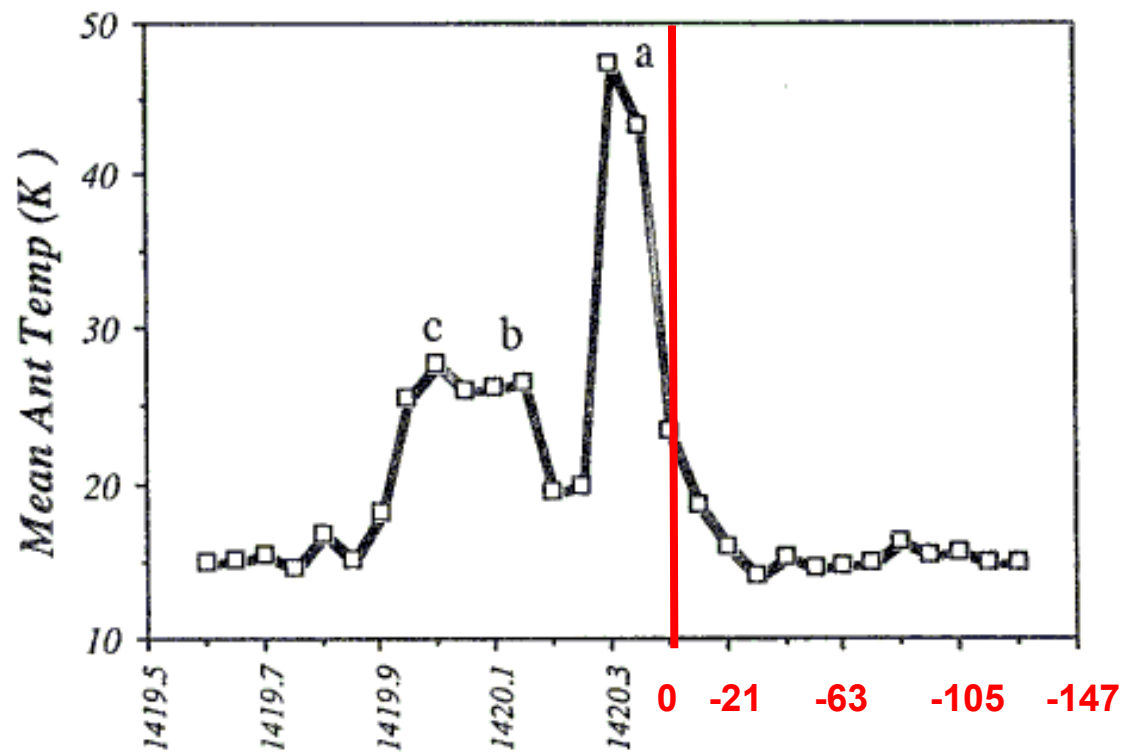
Turning into Velocities

Let's take $\Delta v = 0.1 \text{ MHz} = 1 \text{e}5 \text{ Hz}$ and $v = 1 \text{e}5 \text{ Hz} * 3 \text{e}5 \text{ km/s} / 1420.4 \text{e}6 \text{ Hz} = 21 \text{ km/s}$. This sets our scale....



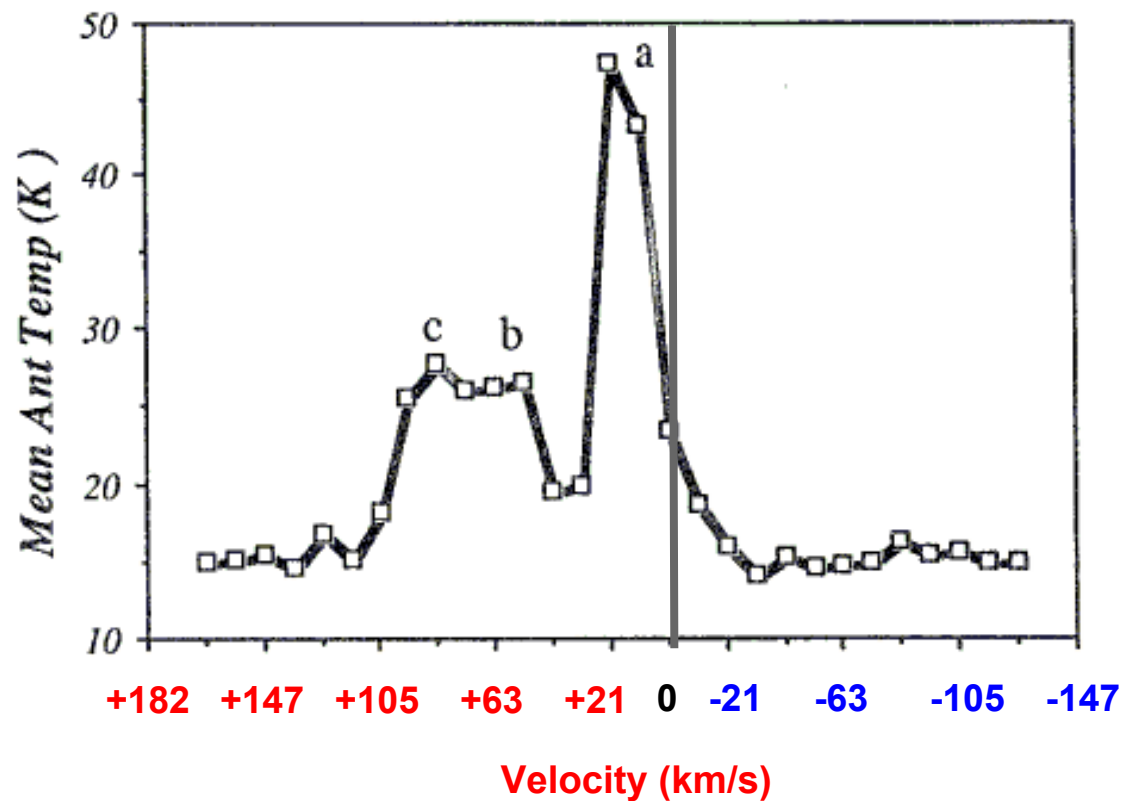
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Correcting to the LSR Velocity Frame

Let's assume our observations were taken now, toward Orion at (RA, Dec) = (+5:35:17, -5:23:28).

<http://www.gb.nrao.edu/GBT/setups/radvelcalc.html>

Correcting to the LSR Velocity Frame

Let's assume our observations were taken now, toward (RA, Dec) = (+19h, +15deg.), or (l, b) = (47.3, 4.9)

<http://www.gb.nrao.edu/GBT/setups/radvelcalc.html>

I get:

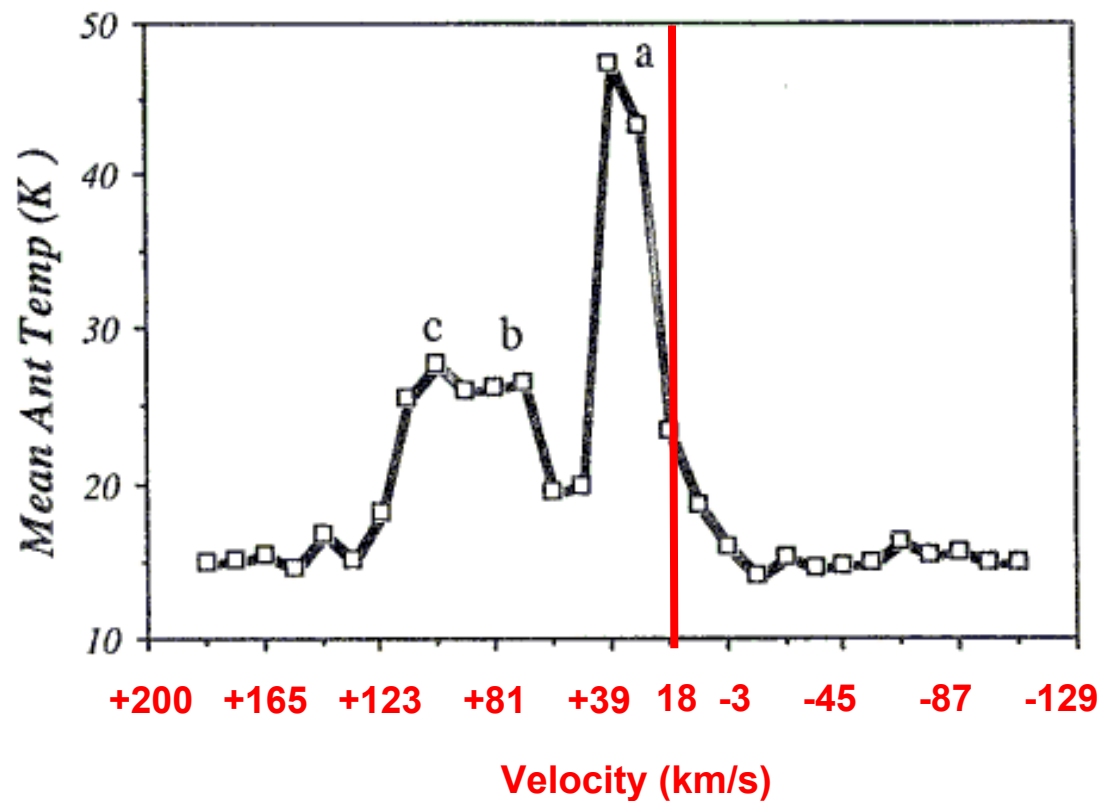
$v_{\text{helio}} = +0.34 \text{ km/s}$

$v_{\text{lsr}} = -18.5 \text{ km/s}$

So the LSR velocity frame is moving toward us at 18.5 km/s. We must add 18.5 km/s to our velocity axis.

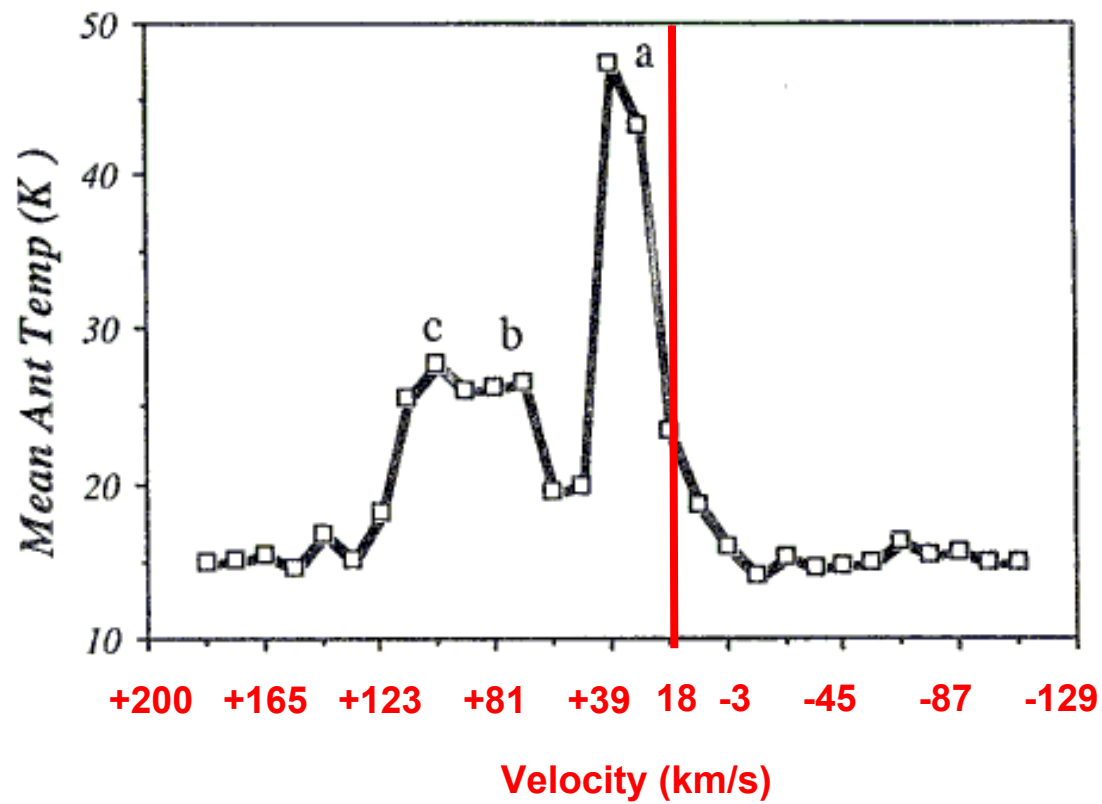
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Results!

We have peaks at:
+39, +70, and +100 km/s



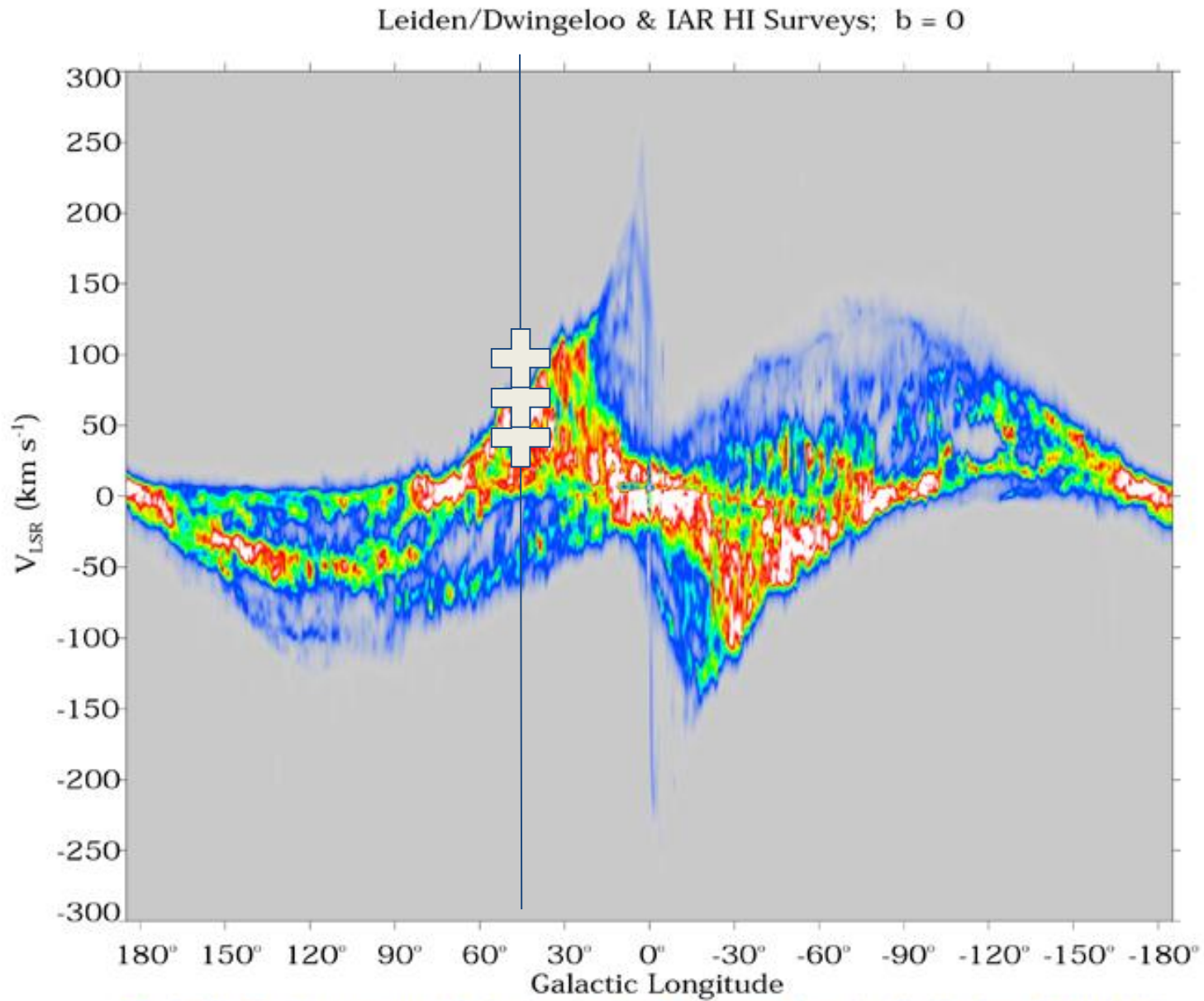


Fig 2.20 (D. Hartmann) 'Galaxies in the Universe' Sparke/Gallagher CUP 2007

Where is this gas?

$$V_{\text{rad}} = (\omega - \omega_0) R_0 \sin l$$

where ω is the angular velocity defined as v/R , R is the distance from the center of the Galaxy, l is the Galactic longitude, and the “0” subscripts denote Solar quantities. $R_0 = 8.3\text{kpc}$ and $\omega_0 = 29 \text{ (km/s)/kpc}$.

Let's assume the rotation curve is completely flat, so $v = 230\text{km/s}$ everywhere. Then, $\omega = 230 \text{ km/s} / R$

Rearranging,

$$\omega = V_{\text{rad}} / (R_0 \sin l) + \omega_0 = 230 \text{ km/s} / R$$

or

$$R = 230 \text{ km/s} / (V_{\text{rad}} / (8.3\text{kpc} \sin l) + 29 \text{ (km/s)} / \text{kpc})$$

Where is this gas?

$$R = 230 \text{ km/s} / (V_{\text{rad}} / (8.3 \text{ kpc} \sin l) + 29 \text{ (km/s)} / \text{kpc})$$

So for $V_{\text{rad}} = +30, +70, +100 \text{ km/s}$ at $l = 47^\circ$,

$R = 6.8, 5.7, 5.1 \text{ kpc}$

